

Multiple Signal Classification (MUSIC) Algorithm

High-resolution Direction-of-Arrival Estimation

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Array Geometry

Array Geometry:

- **Uniform Linear Array (ULA):**

- Number of elements: $N = 5$
- Inter-element spacing: $d = \frac{\lambda}{2}$
- Number of elements: $N = 9$
- Inter-element spacing: $d = \frac{\lambda}{2}$

- **Uniform Circular Array (UCA):**

- Number of elements: $N = 5$
- Angular spacing: $\Delta\gamma = \frac{2\pi}{5}$
- Radius: $r = 0.5$
- Inter-element arc spacing: $r \cdot \frac{2\pi}{5} = 0.2\pi$
- Number of elements: $N = 9$
- Angular spacing: $\Delta\gamma = \frac{2\pi}{9}$
- Radius: $r = 0.8$
- Inter-element arc spacing: $r \cdot \frac{2\pi}{9} \approx 0.178\pi$

Signal Model

For an N element array, with D number of sources impinging on it for T snapshots in time,

Signal Model:

$$\mathbf{X} = \mathbf{A}\mathbf{S} + \mathbf{n}$$

- $\mathbf{X}$: $N \times T$ received data matrix
- $\mathbf{A}$: $N \times D$ steering matrix, with columns $\mathbf{a}(\theta_d)$
- $\mathbf{S}$: $D \times T$ source signal matrix
- $\mathbf{n}$: additive white Gaussian noise

ULA MUSIC Algorithm

Steering Vector:

$$\mathbf{a}(\theta) = [1 \quad e^{jkd \cos \theta} \quad \dots \quad e^{jkd(N-1) \cos \theta}]^T$$

Sample Covariance Matrix:

$$\mathbf{R} = \frac{1}{T} \mathbf{X} \mathbf{X}^H$$

Eigen-Decomposition:

$$\mathbf{R} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^H$$

- Select noise subspace: $\mathbf{V}_n$ (eigenvectors corresponding to smallest $N - D$ eigenvalues)

MUSIC Spectrum:

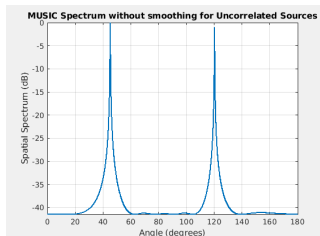
$$P_{\text{MUSIC}}(\theta) = \frac{1}{\mathbf{a}^H(\theta) \mathbf{V}_n \mathbf{V}_n^H \mathbf{a}(\theta)}$$

ULA-MUSIC for Two Non-Coherent Sources

Setup:

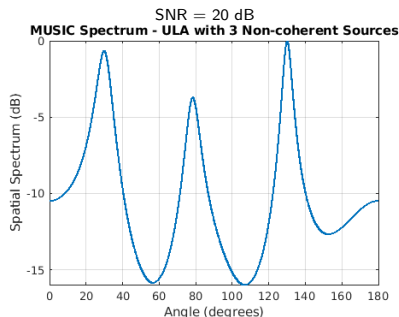
- Two narrowband, unmodulated sinusoidal signals
- True DOAs: $\theta = [45^\circ, 120^\circ]$
- Signal Model:

$$S_d(t) = e^{(j2\pi f_d t)}, \quad d = 1, \dots, D$$



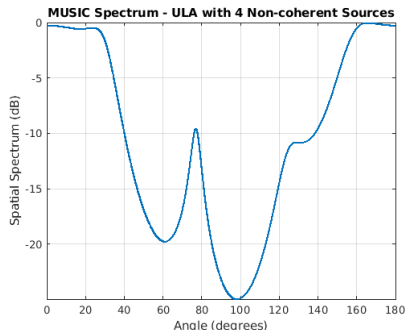
	Source 1	Source 2
True DOAs (°)	45.00	120.00
Estimated DOAs (°)	45.01	120.00
Peak Powers (dB)	0.00	-1.02

ULA-MUSIC for Three Non-Coherent Sources

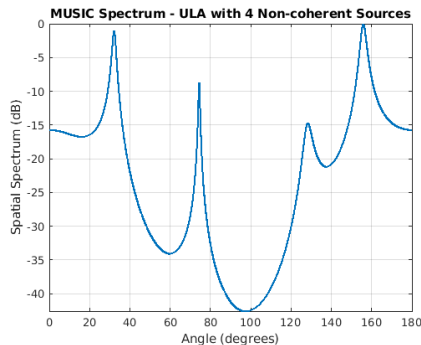


	source 1	source 2	source 3
True DOA (°)	30.00	75.00	130.00
Estimated DOA (°)	29.70	78.19	129.85
Peak Power (dB)	-0.65	-3.70	0.00

ULA-MUSIC for Four Non-Coherent Sources



SNR = 20 dB



SNR = 40 dB

True DOAs	30°	75°	130°	160°
Estimated DOAs	31.97°	74.19°	128.15°	155.67°
Peak Powers (dB)	-1.04	-8.70	-14.71	0

Signal Model – Coherent Sources

Two sources share the same base signal (coherent/multipath case)

Base Signal:

$$s_1(t) = e^{(j2\pi ft)}$$

Reflected (Delayed) Signal:

$$s_2(t) = \begin{cases} 0, & t < \text{delay} \\ s_1(t - \text{delay}), & \text{otherwise} \end{cases}$$

Received Signal at Array:

$$\mathbf{X} = r_1 \mathbf{a}(\theta_1) s_1(t) + r_2 \mathbf{a}(\theta_2) s_2(t)$$

Parameters:

- $f = 5 \times 10^8$ Hz (carrier frequency)
- Delay = 5 samples
- $r_1 = \sqrt{0.6}$, $r_2 = \sqrt{0.4}$
- DOAs: $\theta_1 = 45^\circ$, $\theta_2 = 120^\circ$
- $t = nT_s$, where n = snapshots, $T_s = 1/f_s$, $f_s = 100$ kHz

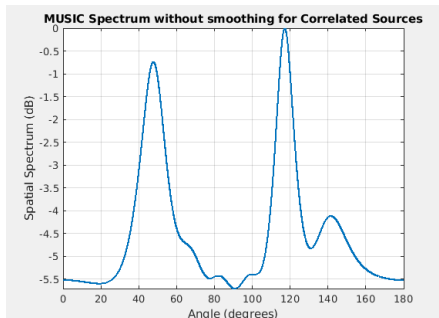
Spatial Smoothing – Resolving Coherent Sources

- Divide the N -element ULA into $L = N - P + 1$ overlapping subarrays of size P .
- Compute local covariance matrices $\mathbf{R}_l$ for each subarray.
- Average them to obtain the smoothed covariance:

$$\mathbf{R}_{\text{smoothed}} = \frac{1}{L} \sum_{l=1}^L \mathbf{R}_l$$

- Reapply MUSIC using the new full-rank $\mathbf{R}_{\text{smoothed}}$ [1].

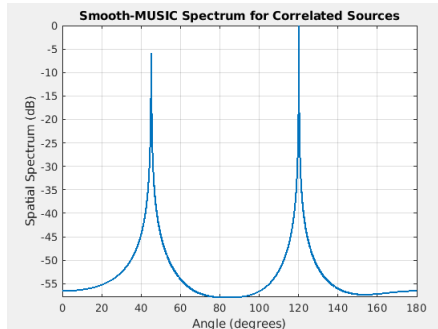
ULA-MUSIC vs Smoothed-ULA-MUSIC for Two Coherent Sources (LOS + Multipath component) in ULA



	Source 1	Source 2
True DOA	45°	120°
Estimated DOA	47.55°	116.96°
Peak Power	-0.73 dB	0 dB

SNR = 20 dB

Unsmoothed MUSIC

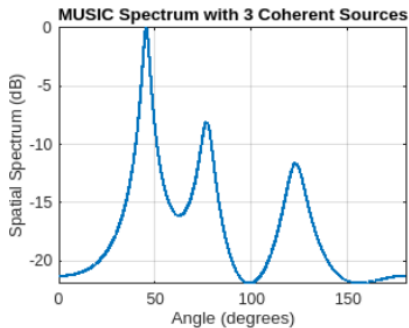


	Source 1	Source 2
True DOA	45°	120°
Estimated DOA	44.96°	119.93°
Peak Power	-5.84 dB	0 dB

SNR = 20 dB

Spatially Smoothed MUSIC (P = 4)

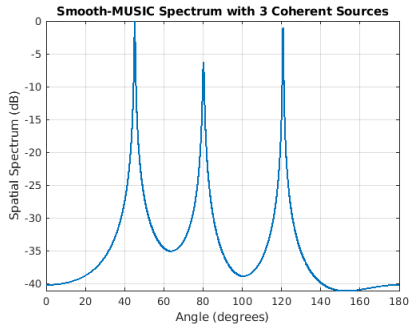
ULA-MUSIC vs Smoothed-ULA-MUSIC for Three Coherent Sources (LOS + Multipath components) in ULA



Unsmoothed MUSIC

SNR = 20dB

	Source 1	Source 2	Source 3
True (°)	45.00	80.00	120.00
Est. (°)	44.76	80.00	120.56
Power (dB)	0.00	-20.21	-22.80



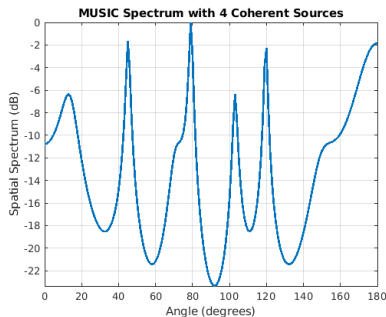
Spatially Smoothed MUSIC

$r = (\sqrt{0.6}, \sqrt{0.3}, \sqrt{0.1})$, $P = 4$

SNR = 20dB

	Source 1	Source 2	Source 3
True (°)	45.00	80.00	120.00
Est. (°)	44.98	80.00	120.56
Power (dB)	0.00	-6.25	-1.04

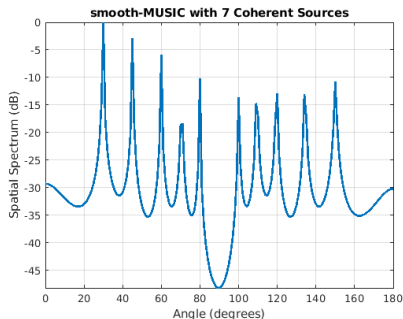
ULA MUSIC for Four Coherent Sources (LOS + Multipath Components)



$$r = (\sqrt{0.4}, \sqrt{0.3}, \sqrt{0.2}, \sqrt{0.1}), \quad N = 5, \quad \text{SNR} = 20 \text{ dB}$$

	Source 1	Source 2	Source 3	Source 4
True DOA (°)	45.00	80.00	120.00	150.00
Est. DOA (°)	45.00	79.00	120.00	103.00
Power (dB)	-1.98	0.00	-2.00	-6.10

ULA MUSIC for Seven Coherent Sources (LOS + Multipath Components)



$$r = (\sqrt{0.4}, \sqrt{0.3}, \sqrt{0.25}, \sqrt{0.2}, \sqrt{0.15}, \sqrt{0.1}, \sqrt{0.05}, \sqrt{0.02}),$$

$$N = 9, P = 8, \quad \text{SNR} = 40 \text{ dB}$$

	Src 1	Src 2	Src 3	Src 4	Src 5	Src 6	Src 7
True DOA (°)	30	45	60	80	100	120	150
Est. DOA (°)	30	45	60	80	100	109	150
Power (dB)	-2.69	-2.12	0.00	-7.14	-10.28	-10.41	-10.33

Limit of Resolution

- **Note:** Due to the constraint on P , spatial smoothing is not possible for $N = 5$ when $D = 4$.
- (For a strict implementation of the algorithm, assuming number of sources are known) The smoothed signal covariance matrix has dimensions $P \times P$, which imposes a lower bound on P :
At least D eigenvalues correspond to the signal subspace, so at least one must remain for the noise subspace.
- **Constraint on subarray size P :**

$$D < P < N$$

- **Resolution Capabilities:**
 - **ULA (without smoothing):** can resolve up to $N - 1$ **independent** sources [2].
 - **With spatial smoothing:** can resolve up to $\frac{2N}{3}$ **coherent** sources [2].

UCA Element Space MUSIC: Theory and Model

Overview:

- UCA: Elements placed uniformly on a circle of radius r
- DOA defined in spherical coordinates (θ, ϕ)
- Steering vector:

$$\mathbf{a}(\theta, \phi) = e^{(jk_0 r \sin \theta \cos(\phi - \gamma_n))}, \quad \gamma_n = \frac{2\pi n}{N}$$

- Covariance matrix $R = \frac{1}{T} \mathbf{X} \mathbf{X}^H$
- MUSIC spectrum:

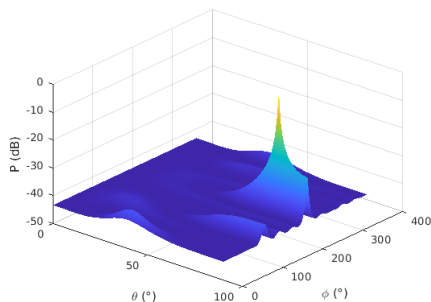
$$P(\theta, \phi) = \frac{1}{\mathbf{a}^H(\theta, \phi) \mathbf{V}_n \mathbf{V}_n^H \mathbf{a}(\theta, \phi)}$$

- Peaks of $P(\theta, \phi)$ correspond to estimated DOAs

Simulation Setup:

- $N = 5$ elements, $T = 200$ snapshots

UCA Element Space MUSIC: Single Source Detection

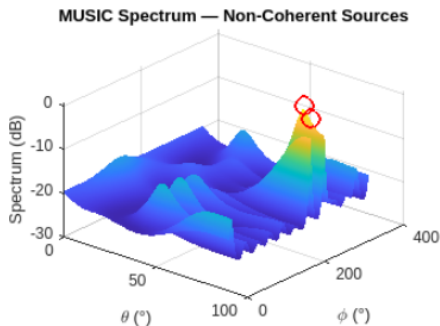


MUSIC for Single Source

True DOA (θ, ϕ)	(75°, 210°)
Estimated DOA	(75°, 210°)
SNR	20 dB

UCA Element Space MUSIC: Multisource Detection

Two Non-Coherent Sources

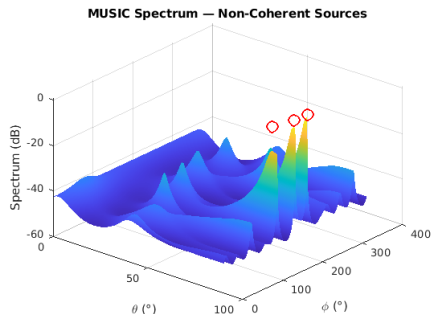


SNR = 20 dB
Elevation (θ) 90°

	source 1	source 2
True DOAs	210°	180°
Estimated DOAs	210°	179°
Peak Powers	0 dB	-0.14 dB

UCA Element Space MUSIC: Multisource Detection

Three Non-Coherent Sources



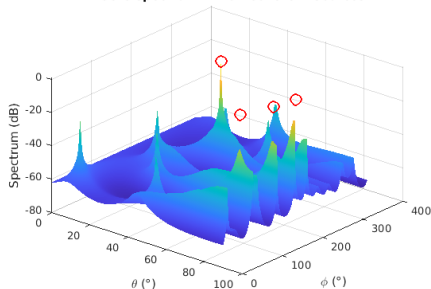
SNR = 20 dB
Elevation (θ) 90°

	source 1	source 2	source 3
True DOAs	210°	180°	135°
Estimated DOAs	210°	180°	135°
Peak Powers	0 dB	-2.2 dB	-9.52 dB

UCA Element Space MUSIC: Multisource Detection

Four Non-Coherent Sources

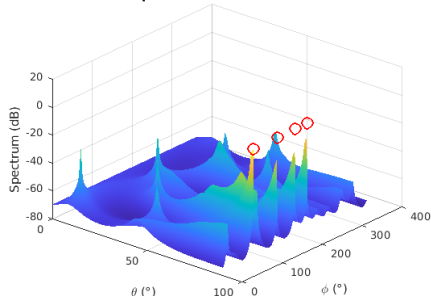
MUSIC Spectrum — Non-Coherent Sources



SNR = 20 dB
Elevation (θ) 90°

	src 1	src 2	src 3	src 4
True	210°	180°	135°	75°
Est.	228°	180°	135°	75°
Powers	0 dB	-11.32 dB	-19.88 dB	-19.60 dB

MUSIC Spectrum — Non-Coherent Sources



SNR = 40 dB
Elevation (θ) 90°

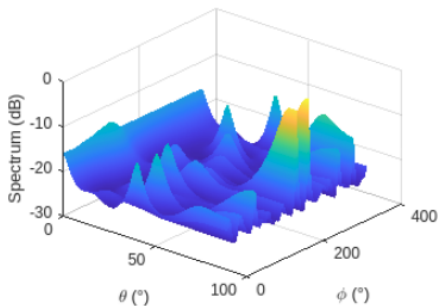
	src 1	src 2	src 3	src 4
True	210°	180°	135°	75°
Est.	210°	180°	135°	75°
Powers	-9.52 dB	-14.84 dB	-14.97 dB	0 dB

Limit of Resolution of UCA element space MUSIC

- UCA element space MUSIC can resolve $N - 1$ non-coherent sources.
- For $N=5$, resolution limit = 4, as indicated by the plots.

UCA Element Space MUSIC: Multisource Detection

Two Coherent Sources (LOS + Reflected)



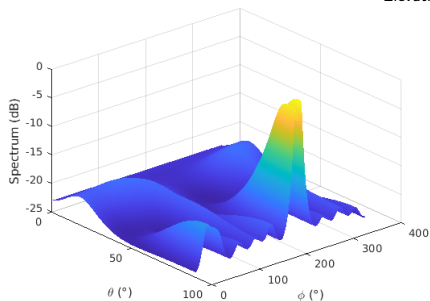
SNR = 20 dB

Elevation (θ) 90°

	source 1	source 2
True	210°	180°
Est.	210°	180°
Powers	0 dB	-1.52 dB

How Close Can the Sources Be? (Limit of Spatial Separation)

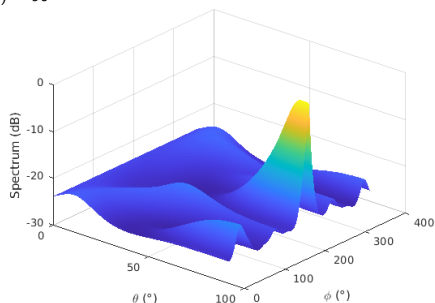
SNR = 20 dB
Elevation (θ) = 90°



10° separation

	Source 1	Source 2
True	210°	220°
Est.	211°	223°
Powers	0 dB	-0.95 dB

Stable localization at 10° separation.



5° separation

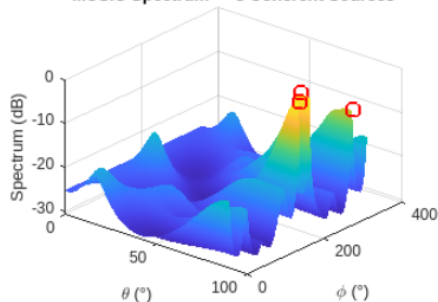
	Source 1	Source 2
True	210°	205°
Est.	208°	38°
Powers	0 dB	-19.35 dB

Degradation below 10° separation evident.

UCA element space MUSIC: Multisource detection — Coherent sources (3 and 4 sources)

SNR = 20 dB
Elevation (θ) 90°

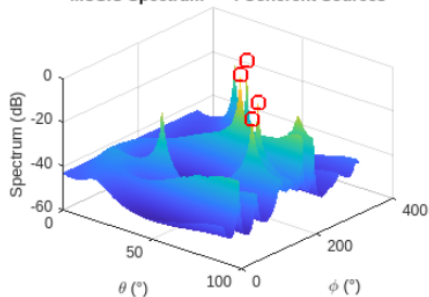
MUSIC Spectrum — 3 Coherent Sources



$r = (\sqrt{0.5}, \sqrt{0.4}, \sqrt{0.1})$
Delay (samples): (0, 5, 8)

	src 1	src 2	src 3
True	210°	180°	320°
Estimated	210°	181°	320°
Powers (dB)	0	-0.14	-7.69

MUSIC Spectrum — 4 Coherent Sources



$r = (\sqrt{0.5}, \sqrt{0.3}, \sqrt{0.2}, \sqrt{0.1})$
Delay (samples): (0, 5, 8, 10)

	src 1	src 2	src 3	src 4
True	210°	180°	150°	120°
Estimated	205°	181°	151°	285°
Powers (dB)	0	-6.68	-10.57	-3.24

Limit of Resolution

- For a **5-element array**, performance degrades drastically beyond **3 sources**.
- For **3 sources**: Among 100 Monte Carlo trials:
 - 16 estimates were within **2°** (azimuth) tolerance.
 - 46 estimates were within **3°** (azimuth) tolerance.
 - 61 estimates were within **4°** (azimuth) tolerance.
- This suggests an empirical resolution limit of:

$$\frac{N - 1}{2}$$

UCA Real Beamspace MUSIC

- Convert UCA element space data into a virtual ULA which yields a real-valued beamspace to simplify MUSIC spectrum computation.

Updated MUSIC spectrum:

$$P(\theta, \phi) = \frac{1}{\mathbf{a}_r^H(\theta, \phi) \mathbf{V}_n \mathbf{V}_n^H \mathbf{a}_r(\theta, \phi)}$$

where $\mathbf{a}_r = \mathbf{F}_r^H \mathbf{a}$ is the real beamspace-transformed steering vector and $\mathbf{F}_r^H$ is the real-valued beamforming matrix.

UCA Real Beamspace MUSIC Overview

Array Parameters:

- N sensors placed in circular array of radius r
- Angular positions $\gamma_n = \frac{2\pi n}{N}$, $n = 0, 1, \dots, N - 1$

Beamspace Transformation:

$$\mathbf{a}_r(\theta, \phi) = \mathbf{F}_r^H \cdot \mathbf{a}(\theta, \phi) = \sqrt{N} \cdot \mathbf{W}^H \cdot \mathbf{J}_\zeta \cdot \mathbf{V}(\phi)$$

Where:

- $\mathbf{F}_r$ is the real-valued beamforming transformation
- $\mathbf{a}(\theta, \phi)$ is the spatial steering vector
- $\mathbf{J}_\zeta$ is diagonal Bessel response
- $\mathbf{V}(\phi)$ is a DFT vector (modal function)
- $\zeta = k_0 r \sin \theta$

Noise Subspace Projection in Beamspace:

$$P(\theta, \phi) \propto \frac{1}{\mathbf{V}^H(\phi) \mathbf{J}(\zeta) (\mathbf{W} \mathbf{V}_n \mathbf{V}_n^T \mathbf{W}^H) \mathbf{J}(\zeta) \mathbf{V}_r(\phi)}$$

Let:

$$\mathbf{Q}_\zeta = \mathbf{J}_\zeta \cdot \mathbf{W} \cdot \mathbf{V}_n \mathbf{V}_n^T \cdot \mathbf{W}^H \cdot \mathbf{J}_\zeta$$

The denominator in $\mathbf{P}(\theta, \phi)$ is the **Null Spectrum**:

$$P_{\text{Null}}(\phi) = \sum_{\ell=-(M'-1)}^{M'-1} a_{\zeta}(\ell) \cdot e^{j\ell\phi}$$

Each coefficient $a_{\zeta}(\ell)$ is obtained from the diagonals of the matrix $\mathbf{Q}_{\zeta}$:

$$a_{\zeta}(\ell) = \sum_{\substack{i,j \\ j=i+\ell}} [\mathbf{Q}_{\zeta}]_{i,j}$$

Due to symmetry of $\mathbf{Q}_{\zeta}$, define the symmetric sequence:

$$a'_{\zeta} = \{a_{\zeta}(0), 2a_{\zeta}(1), \dots, 2a_{\zeta}(M' - 1)\}$$

Then the null spectrum becomes:

$$P_{\text{Null}}(\phi) = \sum_{\ell=0}^{M'-1} a'_{\zeta}(\ell) \cdot e^{-j\ell\phi}$$

This is recognized as a Discrete Fourier Transform:

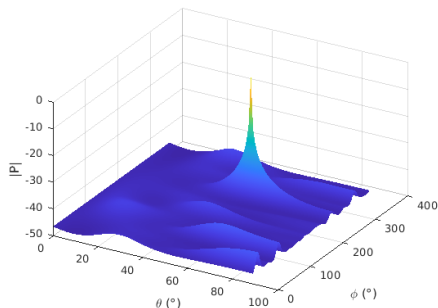
$$P_{\text{Null}} = \text{FFT} (a'_{\zeta}) [l], \quad \phi_k = \frac{2\pi l}{L}, \quad l = 0, 1, \dots, L-1$$

Thus, the spectrum over azimuth is computed efficiently via the FFT of a'_{ζ} .

$\mathbf{Q}_{\zeta}$ depends only on elevation (θ), so 90 FFTs are required to carry out the complete spectral search.

Size of FFT in simulations: $L = 4096$

Performance: Real Beamspace MUSIC (Single Source)



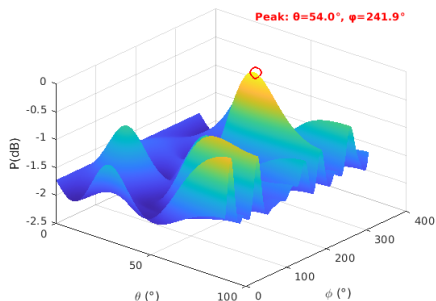
Without FFT Optimization

True DOA	$(53^\circ, 243^\circ)$
Estimated DOA	$(53^\circ, 243^\circ)$

SNR = 20 dB

Source frequency = 600 MHz

Time: 0.161 s



With FFT Optimization

True DOA	$(53^\circ, 243^\circ)$
Estimated DOA	$(54^\circ, 241.9^\circ)$

SNR = 20 dB

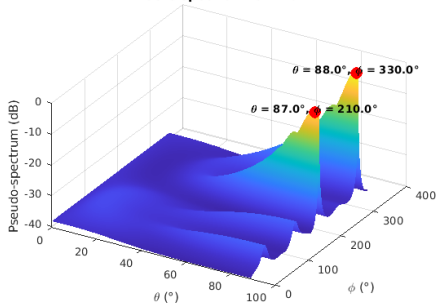
Source frequency = 600 MHz

Time: 0.022 s

UCA-RB-MUSIC for Multiple Sources: Two Non-coherent sources

SNR = 30 dB
Elevation (θ) 90°

MUSIC Spectrum for Trial #2



Without FFT Optimization

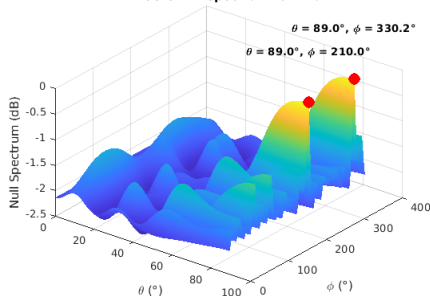
	Source 1	Source 2
True	210°	330°
Est.	210°	330°
Powers	0 dB	-2.24 dB

SNR = 30 dB

Time = 13.08 s (100 trials)

Note: For 100 Monte Carlo trials, 99 estimates were accurate within 1° error.

RB-MUSIC FFT Spectrum for Trial #2



With FFT Optimization

	Source 1	Source 2
True	210°	330°
Est.	210°	330.2°
Powers	0 dB	0 dB

SNR = 30 dB

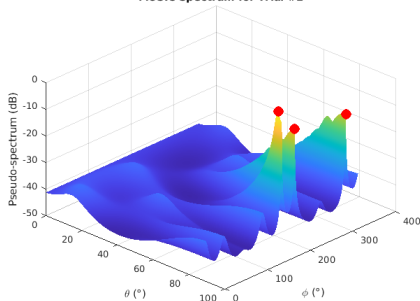
Time = 4.4 s (100 trials)

Note: All 100 trials yield correct estimates within 1° error.

UCA-RB-MUSIC for Multiple Sources: Three Non-coherent sources

SNR = 30 dB
Elevation (θ) = 90°

MUSIC Spectrum for Trial #2



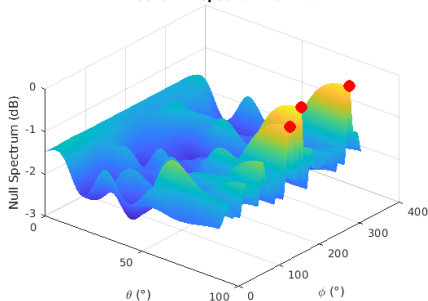
Without FFT Optimization

	Source 1	Source 2	Source 3
True	210°	330°	180°
Est.	210°	330°	180°
Powers	-8.25 dB	-11.36 dB	0 dB

SNR = 30 dB
Time = 13.08 s (100 trials)

Note: For 100 Monte Carlo trials, 99 estimates were accurate within 1° error.

RB-MUSIC FFT Spectrum for Trial #2



With FFT Optimization

	Source 1	Source 2	Source 3
True	210°	330°	180°
Est.	209.3°	330.5°	182.5°
Powers	0 dB	-0.09 dB	-0.32 dB

SNR = 30 dB
Time = 4.47 s (100 trials)

Note: For 100 Monte Carlo trials 62 yield correct estimates within 1° error.

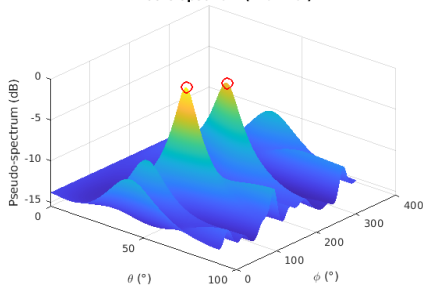
Limit of Resolution for Non-Coherent Sources

- For both UCA-RB-MUSIC and UCA-RB-MUSIC FFT, the limit is $\lceil \frac{N}{2} \rceil$

UCA-RB-MUSIC for Multiple Sources: Two Coherent Sources (LOS + Reflected Multipath)

SNR = 30 dB
Elevation (θ) 45°

MUSIC Spectrum (2nd Trial)



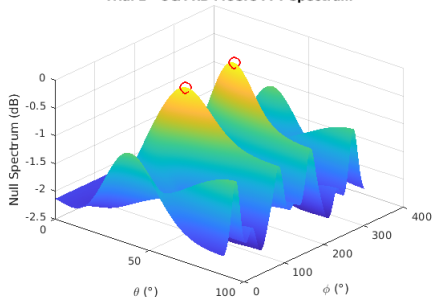
Without FFT Optimization

	Source 1	Source 2
True	$(45^\circ, 120^\circ)$	$(45^\circ, 245^\circ)$
Est.	$(47^\circ, 120^\circ)$	$(46^\circ, 248^\circ)$
Powers	0 dB	-0.78 dB

Time = 12.91 s (100 trials)

Note: 87 trials yielded estimates within 2° error.

Trial 2 - UCA RB-MUSIC FFT Spectrum



With FFT Optimization

	Source 1	Source 2
True	$(45^\circ, 120^\circ)$	$(45^\circ, 245^\circ)$
Est.	$(45^\circ, 119.91^\circ)$	$(45^\circ, 246.33^\circ)$
Powers	-0.04 dB	0 dB

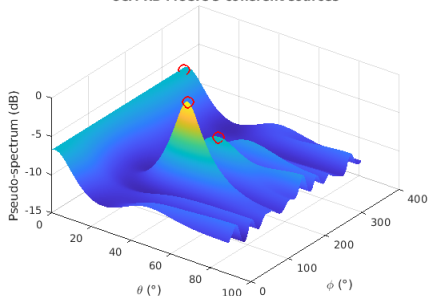
Time = 3.47 s (100 trials)

Note: 98 trials yielded estimates within 2° error.

UCA-RB-MUSIC for Multiple Sources: Three Coherent Sources (LOS + Reflections)

SNR = 30 dB
 $\alpha = 45^\circ$

UCA-RB-MUSIC 3 coherent sources



Without FFT Optimization

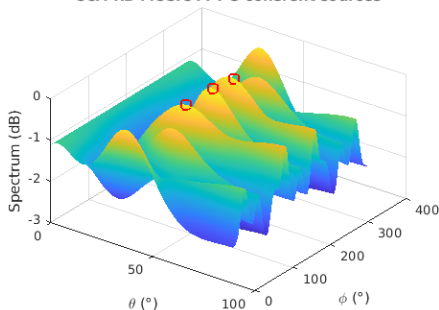
	Src 1	Src 2	Src 3
True	120°	180°	240°
Est.	121°	182°	348°
Powers (dB)	0 dB	-6 dB	-6.6 dB

$r = (\sqrt{0.6}, \sqrt{0.3}, \sqrt{0.1})$, delays = (0, 5, 8)

Time = 15.46 s (100 trials)

For trials, none yield estimates for all three sources within 1° error. 62/100 for only LOS estimates.

UCA-RB-MUSIC FFT 3 coherent sources



With FFT Optimization

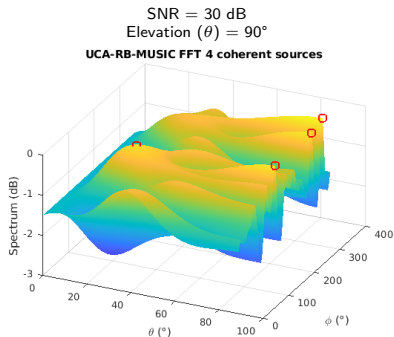
	Src 1	Src 2	Src 3
True	120°	180°	240°
Est.	119°	180°	240°
Powers	-0.08 dB	0 dB	-0.09 dB

$r = (\sqrt{0.6}, \sqrt{0.3}, \sqrt{0.1})$, delays = (0, 5, 8)

Time = 1.67 s (100 trials)

46/100 trials within 1° error

UCA-RB-MUSIC for Multiple Sources: Four Coherent Sources (LOS + Reflections)



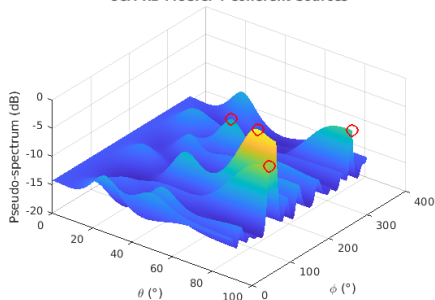
	Source 1	Source 2	Source 3	Source 4
True	120°	180°	240°	320°
Est.	132°	142°	285°	327°
Powers	-0.22 dB	-0.13 dB	-0.12 dB	0 dB

Parameters: $r = (\sqrt{0.4}, \sqrt{0.3}, \sqrt{0.2}, \sqrt{0.1})$
delays = (0, 5, 8, 10)

UCA-RB-MUSIC for Multiple Sources: Four Coherent Sources (LOS + Reflections, 9-element Array)

SNR = 30 dB
Elevation (θ) = 90°

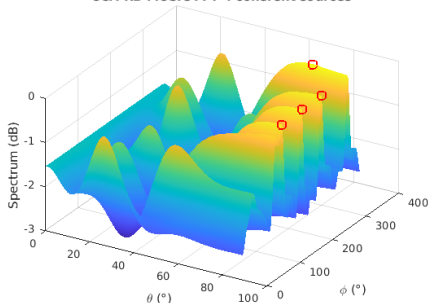
UCA-RB-MUSIC: 4 Coherent Sources



	Src 1	Src 2	Src 3	Src 4
True (Az)	120°	180°	240°	320°
Est. (Az)	120°	102°	252°	318°
Power (dB)	0	-4.62	-8.6	-7.17

Parameters: $r = (\sqrt{0.4}, \sqrt{0.3}, \sqrt{0.2}, \sqrt{0.1})$
delays = (0, 5, 8),
Time: 21.27 s (100 trials)

UCA-RB-MUSIC FFT 4 coherent sources



	Src 1	Src 2	Src 3	Src 4
True (Az)	120°	180°	240°	320°
Est. (Az)	119°	180°	240°	322°
Power (dB)	-0.11	-0.083	-0.087	0

Parameters: $r = (\sqrt{0.4}, \sqrt{0.3}, \sqrt{0.2}, \sqrt{0.1})$
delays = (0, 5, 8, 10),
Time: 1.76 s (100 trials)

Limit of Resolution

- For a **5-element array**, performance degrades drastically beyond **3 sources**.
- For **3 sources**: Among 100 Monte Carlo trials (at SNR = 30 dB):
 - Note: None of the trials yielded correct estimates within 1° error for UCA-RB-MUSIC.
 - Note: 46 out of 100 trials yielded correct estimates within 1° error for UCA-RB-MUSIC with FFT.
- This suggests an empirical resolution limit (at lower SNRs) of:
 - for UCA-RB-MUSIC: 2 (Due to inherent Forward-Backward averaging)
 - for UCA-RB-MUSIC FFT: $\frac{2N}{3}$
- At higher SNRs (> 50 dB), both limits converge to $\frac{2N}{3}$

Summary

Algorithm	Coherent	Non-Coherent
UCA elementspace	$\frac{N-1}{2}$	$N - 1$
UCA-RB-MUSIC	$\frac{2N}{3}$	$\lceil \frac{N}{2} \rceil$

Table: Summary of Source Resolution Limits

Algorithm	Coherent	Non-Coherent
UCA elementspace	0	$N - 1$
UCA-RB-MUSIC	$\frac{2N}{3}$	$\frac{N-1}{2}$

Table: Old Resolution Limits

References I

-  J. C. Liberti and T. S. Rappaport, *Smart Antennas for Wireless Communications: IS-95 and Third Generation CDMA Applications*. Upper Saddle River, New Jersey: Prentice Hall PTR, 1997.
-  Y. Bresler and A. Macovski, "On the number of signals resolvable by a uniform linear array," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 34, pp. 1361–1375, Dec. 1986.